Shock encounters in a forced supersonic box

Native fluid-following Monte Carlo particles retain repeated numerical shock-region encounters, timing brackets, strength evidence and censored outcomes. This report showcases actual local simulation data; it is not a certified physical shock catalogue.

Case	Complete	Open	Complete / particle	Final rms Mach
baseline	57013	1212	6.960	1.316
half_cfl	60813	1169	7.423	1.317
coarse32	35045	1198	4.278	1.263
coarse32_matched	44472	1529	5.429	1.263

Experiment

Periodic unit cube,64^3 cells,8,192 mass-seeded native MC particles; uniform initial density 1 and pressure 1/1.4; initial Fourier velocity rms Mach 5. Newtonian ideal gas gamma=1.4, RK2/PLM/HLLC/CFL.30; continuous native edot=50 forcing, modes 1..3, npeak=2, solenoidal fraction 0.5, OU correlation=0.1, update=0.01. End time 0.30 equals three initial driving-scale turnover times (Ldrive=0.5, initial vrms=5); the later flow evolves and this is not a stationarity assertion.

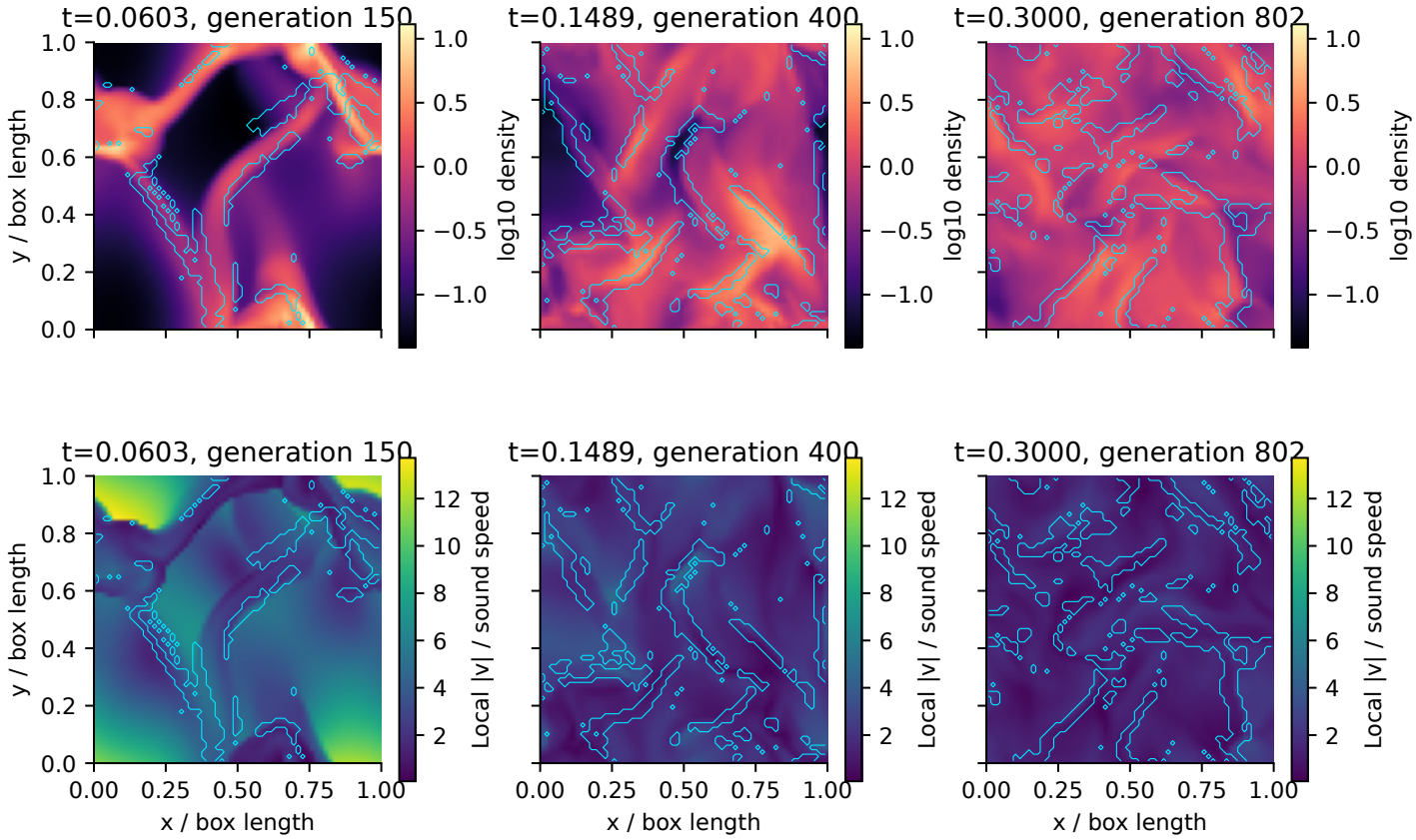
What is checked

Five simulations succeeded. Tracking-on/off entire particle thermo bytes and every saved primitive/conserved/force array agree exactly. All particle IDs survive; event IDs are unique and sequences contiguous; tracker counters agree with strict event readers. Periodic mass error is at most1e-12. Three saved volumes are evaluated through the unchanged native detector fixture with genuine periodic neighbors. Binary times are rounded; plotting and sampling align by generation and full-double native particle clocks.

Paper-use boundary

Use the separate vector PDF/SVG figures and CSV data as a reproducible methods/implementation demonstration. Native RK2 forcing has a separately measured gas-heating bias. There is no cooling, stationary ensemble, accepted strength error budget, physical passage truth catalogue, independent-seed uncertainty estimate or cluster qualification. Captions state these limits next to each figure. All flags and open episodes remain in the data.

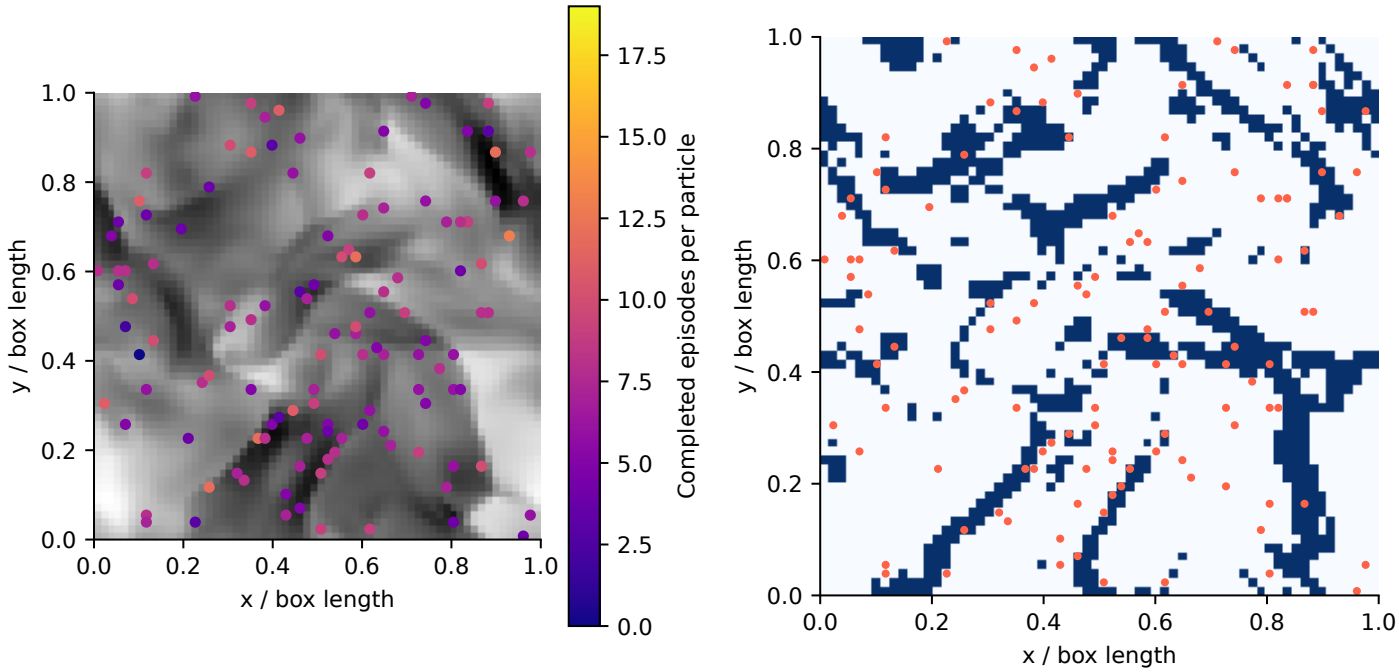
01. Forced supersonic flow and shock regions



Actual z=0.5078125 slices of the 64³ baseline. Colors show log10 density (top) and local speed/sound speed (bottom); each row uses shared color limits. Cyan outlines are the unreduced native detector region recomputed from the exact saved primitive state, with true periodic neighbors, width 4 and min_mach=1.1. The three generations are the available saved states nearest t=.05,.15,.30. Local flow Mach is distinct from detector shock-Mach estimates. Masks demonstrate implemented numerical membership, not a certified physical shock catalogue.

Data: 01_flow.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

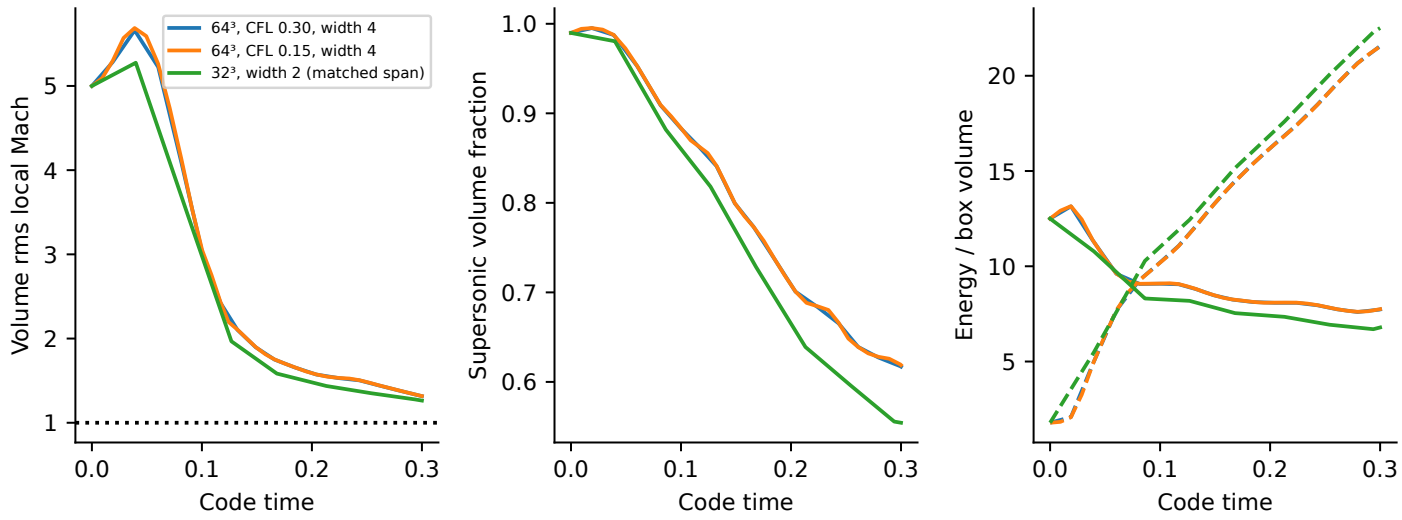
02. Particle histories over the final flow



Only the 123 actual particles whose final parent lies in the plotted z-cell are shown, avoiding a projection of unrelated depths. Left: final density and per-particle accumulated completed-episode count. Right: recomputed native region and the same particles. Counts include their whole recorded history and are not instantaneous shock membership. All 8,192 particle IDs survive; cell-center MC samples are not smooth material trajectories.

Data: 02_particles.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

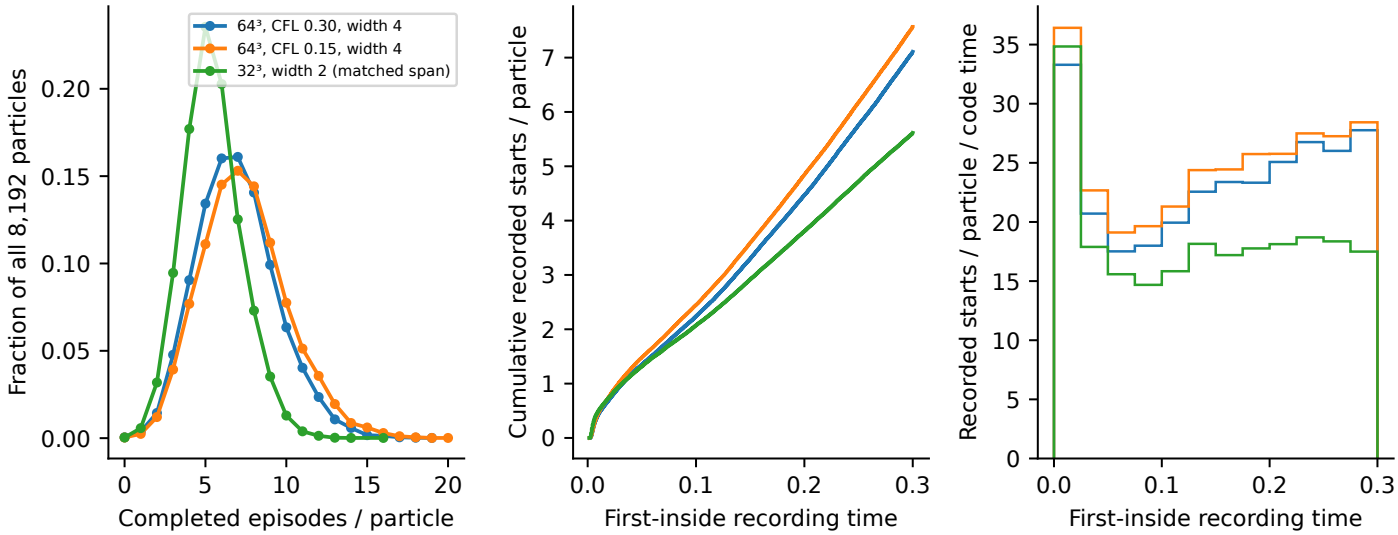
03. Supersonic evolution and heating



Volume rms local Mach is sqrt(mean(|v|²/(gamma P/rho))); supersonic volume fraction counts cells with local Mach>1. Energy panel: solid kinetic, dashed gas internal energy, using the same case colors. Continuous edot=50 forcing is present, but adiabatic shock/source heating increases sound speed. All saved baseline rms Mach values exceed 1; this is an evolving forced realization, not a stationary turbulence ensemble. Total energy includes actual forcing work and the independently measured inherited RK2 source bias; edot is an input normalization target, not a validated physical injection rate.

Data: 03_evolution.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

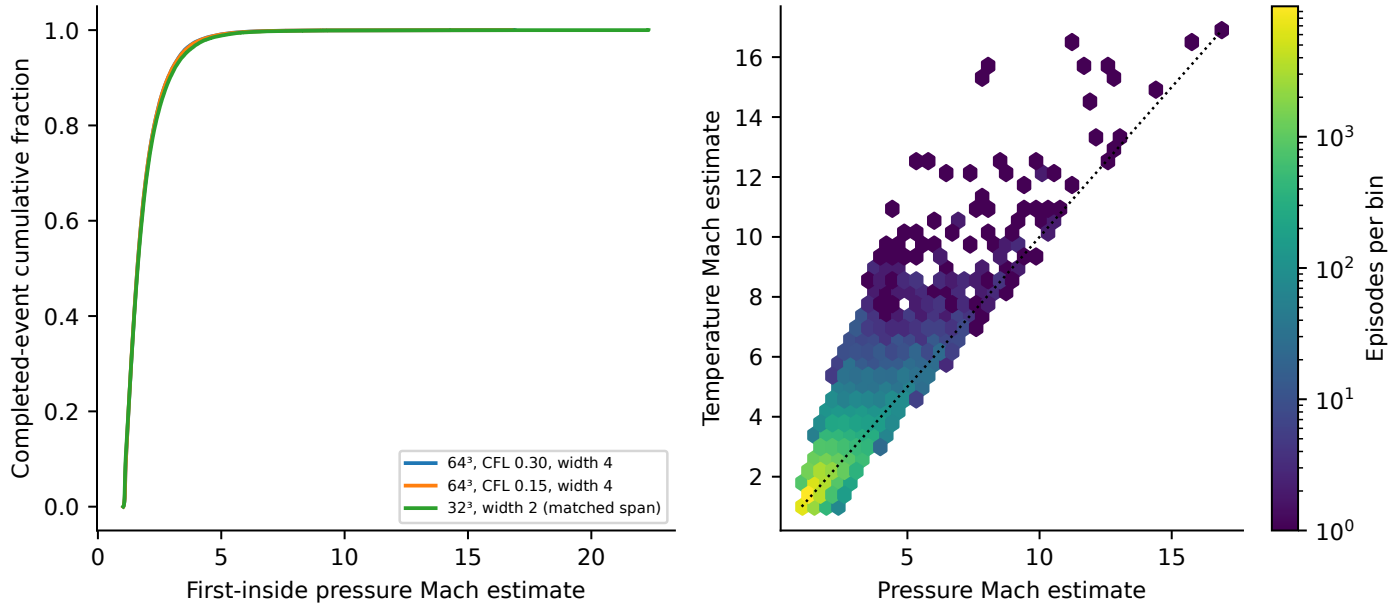
04. How often particles meet detected regions



The count distribution includes all 8,192 particles, including zero-encounter particles. Cumulative/rate panels count each canonical committed or terminal-open identity once at its first-inside observation; left-censored records, if present, are listed separately in the flags table. Rate bins have width 0.025 code time. These are sampled numerical episode recording rates, not exact physical shock-crossing rates; correlated particles and merged flicker prevent a naive independent-event error model. The coarse matched-span case uses width 2 at 32³.

Data: 04_counts.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

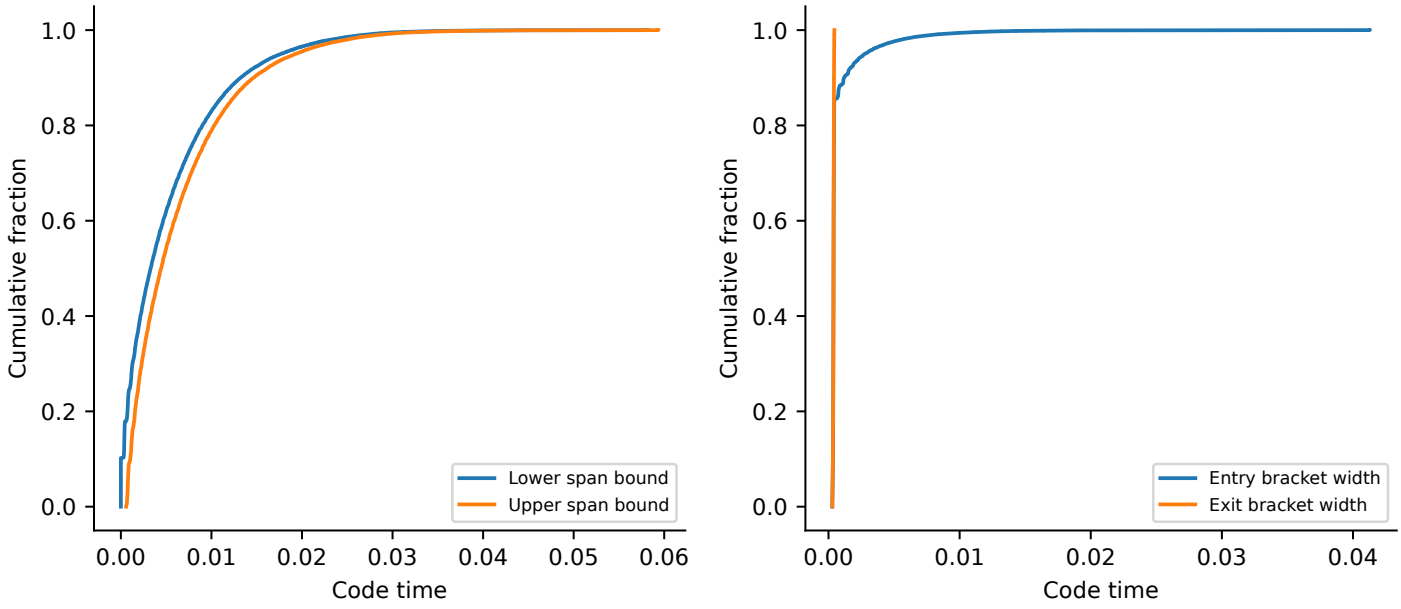
05. Recorded strength evidence



Left: finite first-inside pressure-Mach estimates for completed episodes, normalized per case. Right: baseline pressure/temperature estimators for 57013jointly finite events; 0events without both finite estimators remain in counts/CSV and validity flags. The diagonal is estimator agreement, not an independent physical truth test. Fixed endpoints at±width along the dominant normal axis have no certified plateaus. Either estimator can accept membership; a pressure estimate below 1.1 is therefore possible even with min_mach=1.1.

Data: 05_strength.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

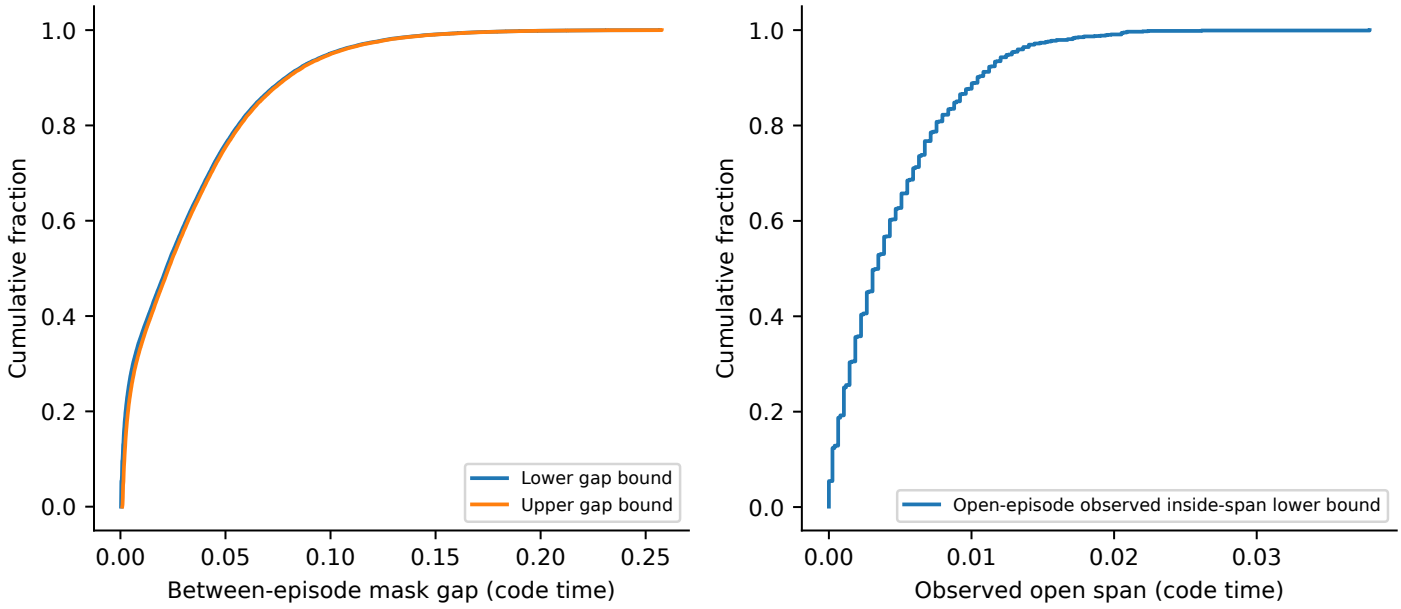
06. Episode spans retain their timing brackets



For uncensored completed episodes, entry lies in[pre,first]and exit in[last,post]. The allowed elapsed-span bounds are[last-first,post-pre]; both endpoint CDFs are shown without inventing midpoints or exact crossing times. Right: widths of the underlying entry/exit observation brackets. Merged flicker can span brief outside intervals, so elapsed span is not continuous residence time. Cadence limitation and stochastic cell hopping remain explicit; these curves are not physical timing-error distributions.

Data: 06_brackets.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

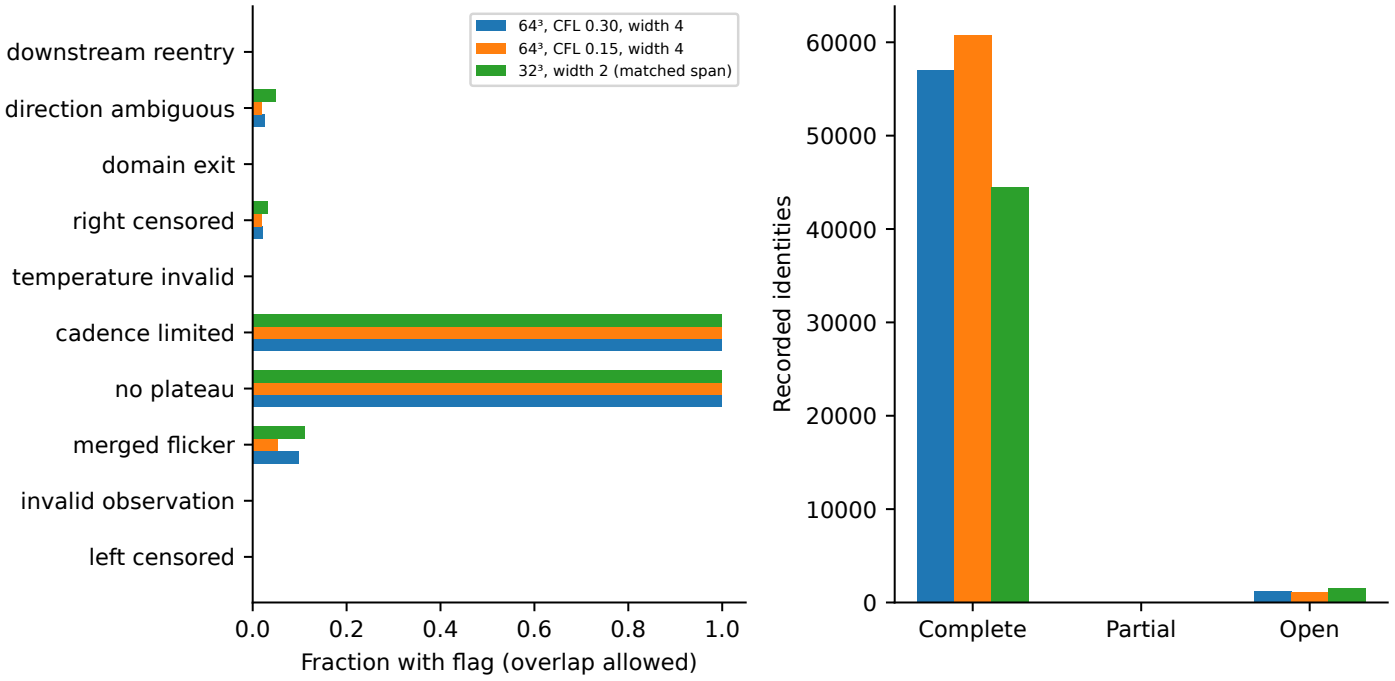
07. Recurrence and right censoring



Left: 48825pairs of consecutive uncensored completed sequences for the same particle. Gap bounds are[max(0,next.pre-previous.post),max(0,next.first-previous.last)]. A zero lower bound means overlapping evidence brackets, not a known zero waiting time. Right: 1212terminal-open identities retained separately, with observed inside-span lower bounds; no finite completed duration is assigned. No stationary waiting-time law, survival fit or independent-episode assumption is claimed.

Data: 07_recurrence.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

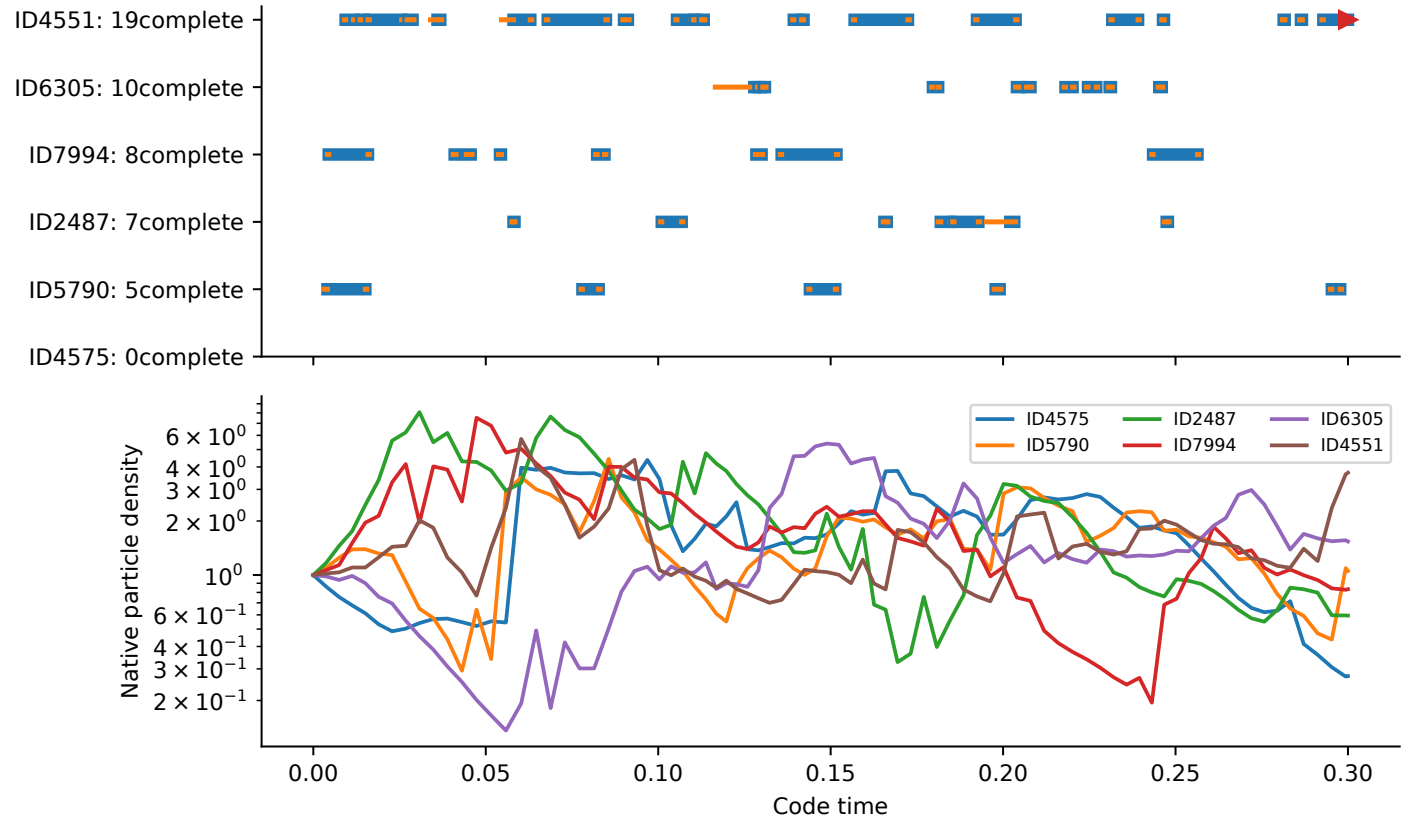
08. Validity flags and censored outcomes



Every event flag is shown, including universal no_plateau/cadence_limited; overlapping flags do not form a probability partition. Direction ambiguity, temperature invalidity and merged flicker remain evidence, not filtered failures. The zero downstream_reentry event-flag bar does not imply zero suppression: the separate surviving-state counter records 75,455 suppressed-entry attempts in the baseline (84,032 at half CFL). These counters are not extra encounter identities. Open terminal views retain canonical identity and can resume after restart; they are not counted as a second completed event. Complete means the numerical tracker obtained its closure evidence, not human certification of a physical passage.

Data: 08_quality.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

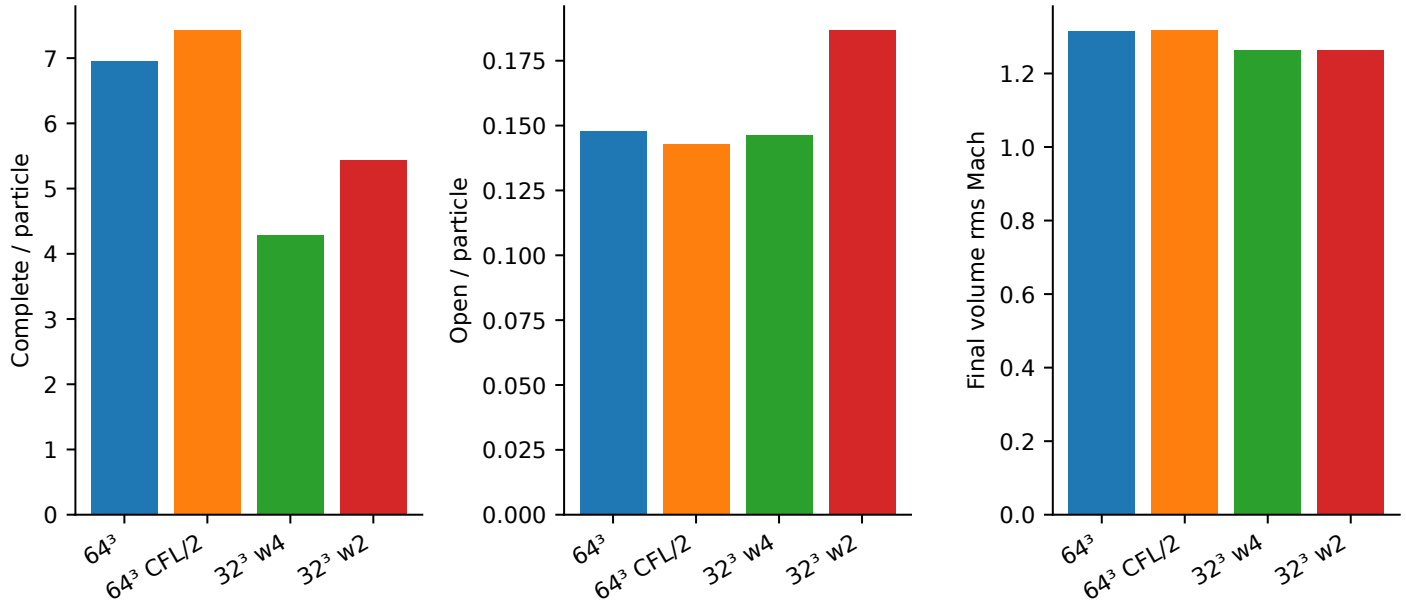
09. Individual particle records without invented trajectories



Particles are selected deterministically at count ranks 0, 25, 50, 75, 90 and 100 percent of the full population, breaking ties by ID; their IDs/counts are printed. Blue segments are first-to-last inside evidence; orange marks entry/exit timing brackets. Red triangles identify open terminal episodes. The lower panel shows actual sampled native density histories; connecting samples is a visual guide, not a continuous material path. Count selection is illustrative rather than an unbiased ensemble sample.

Data: 09_histories.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

10. Timestep, grid and detector-support sensitivity



64³ CFL/2 changes fluid integration, MC draws and observation cadence together.32³ width 4 also doubles the detector physical half-span;32³ width 2 matches the 64³ width 4 half-span(1/16 of box length). The two 32³ runs have byte-identical native particle histories and exact saved fluid/forcing arrays, isolating their detector-width difference. Grid comparisons still mix fluid resolution, stochastic transport and detection. These are measured sensitivities, not convergence acceptance or independent uncertainty estimates. No tolerances were tuned to reduce differences.

Data: 10_sensitivity.pdf / .svg / .png; episodes.csv, evolution.csv and report.json.

Methods, definitions and reproduction

Event and time definitions

Membership uses the unreduced native ShockIt-compatible shock region, min_mach=1.1 and width 4 at 64^3. Temperature gradients orient nearest-axis endpoints; first valid inside evidence is the representative strength. Latest valid outside pre-entry and first inside observations bracket entry; last inside and first valid post observations bracket exit. Two consecutive valid outside observations close/rearm; brief reentries merge and are flagged. Native MC motion follows density fluxes through stochastic parent-cell hops, not integrated smooth trajectories.

Statistics and censoring

Counts include every particle and every unique canonical identity. Completed duration/span bounds are[last-first,post-pre]. Consecutive uncensored completed gaps are[max(0,next.pre-previous.post),max(0,next.first-previous.last)]. No midpoint/exact physical time is fabricated. Terminal-open views are separately retained; they do not contribute finite complete durations. All no_plateau/cadence_limited and other flags are available in the CSV/report. Rates are first-inside recording-time rates, not physical crossing rates.

Comparison interpretation

HalfCFL changes fluid evolution, RNG draws and observation cadence.32^3 width 4 doubles physical endpoint half-span;32^3 width 2 matches64^3 width 4 half-span. Both 32^3 runs have identical native fluid/particle data, isolating their width effect. A single same-seed comparison is sensitivity evidence, not physical convergence or uncertainty. Source bias may affect turbulent temperature/strengths and is not equated with conservative shock heating.

Exact code and environment

Feature coread4225edea78d255b67df97fe8f593c0c268fa1e; compiled fingerprintf6195adafb7e6a554c23a7537053bb00e6ae8c618a10ff1951a487d6981cb70b. Serial CPU/double macOS local run; no production source or diagnostic thresholds were changed to make the plots. Execution/report JSON bind exact executable, commands, source/configuration and logs. Raw simulation path is recorded in README. Parent forcing qualification retains its failing independent thermodynamic oracle.

Reproduce

Run the showcase script with --executable build-release/src/athena --detector build-release/shock_detector_test --workdir /tmp/showcase-new --output /tmp/showcase-plots-new. Both directories must be new. The existing NumPy/Matplotlib environment generates10 figure families, each PDF/SVG/PNG. Run supersonic_showcase_pdf.py on that packet using bundled ReportLab/pypdf Python to generate this report. The README gives exact paths and commands. Production-scale/RUSTY/Frontier and human scientific approval remain separate gates.